# Detailed Methodology for Section 18: Annual Climate and Year-Round Operating Layer

April 19, 2026

## 1 Detailed Methodology for Section 18: Annual Climate and Year-Round Operating Layer

This section documents, in report form, how I implemented the annual climate and daily duty simulation described by Section 18 (229–251). I wrote this layer to convert monthly climate statistics into a physically consistent daily operating profile, then compute daily heating or cooling duty, bed temperature response, electricity use, cost, and water replacement.

**Important modeling clarification (Gaussian terminology).** Section 18 uses periodic Gaussian-kernel regression (weighted averaging), not Gaussian Process Regression (GPR). In this implementation, no posterior covariance model is solved and no Bayesian kernel matrix inversion is performed. Instead, each day is computed as a distance-weighted average of monthly reference values using a Gaussian weighting function, with circular (wraparound) day distance.

### Notation identity clarifications used in Section 18

To avoid symbol ambiguity, I use the following identities throughout this section:

- $m$: month index ($m = 1, \ldots, 12$).
- $d$: day-of-year evaluation index ($d = 1, \ldots, P$).
- $P$: annual period length in days ($P = 365$ in this implementation).
- $d_i$: day-of-year reference day associated with reference point $i$ (month midpoint in this implementation).
- $d_m$: month-midpoint day-of-year for month $m$. With one reference point per month, $d_m$ and $d_i$ refer to the same reference-day map.
- $L_m$: number of calendar days in month $m$.
- $i$ in Eqs. (229)–(230): reference-point index over the monthly climate reference values.
- $i$ in Eqs. (232), (236): day-of-month candidate index inside one month for event placement. This is a different role than $i$ in Eqs. (229)–(230), despite sharing the same symbol.
- $y_i$: reference value at reference point $i$. In Section 18, this is used in two separate passes:

- pass A: $y_i = \bar{T}_i$ (monthly mean temperature),
- pass B: $y_i = \sigma_i$ (monthly standard deviation).

- $\hat{y}(d)$: kernel-regressed daily estimate generated from one selected $y_i$ pass at day $d$.
- $w_i(d)$: Gaussian kernel weight applied to reference point $i$ when evaluating day $d$.
- $h$: Gaussian kernel bandwidth in days.
- $\mathrm{LR}(\cdot)$: largest-remainder integer-allocation operator used for monthly event counts.
- $s_{f,m}$, $r_{f,m}$: freeze-day Gaussian spread and repulsion length scales for month $m$.
- $\mathcal{S}_{f,m}$: set of freeze-day indices already selected in month $m$.
- $\mathcal{S}_{f,m}^{(k)}$: freeze-day selected set after $k$ recursive selections ($\mathcal{S}_{f,m}^{(0)} = \varnothing$).
- $\mathcal{I}_{f,m}^{(k)}$: freeze-day candidates remaining at step $k$, defined by $\mathcal{I}_{f,m}^{(k)} = \{1, \ldots, L_m\} \setminus \mathcal{S}_{f,m}^{(k-1)}$.
- $\mathcal{I}$ is a calligraphic capital $I$ (candidate-index set), not $L$.
- $i_{f,m,k}$: freeze day selected at recursion step $k$.
- $\mathrm{score}_{f,m}^{(k)}(i)$: deterministic ranking objective for freeze-day candidate $i$ at step $k$; larger score means better candidate for selection.
- $M_{f,m}$: freeze-anomaly monthly center value for month $m$ (mean freeze anomaly below $\bar{T}_m$).
- $\Delta T_{f,m}^{\mathrm{low}}, \Delta T_{f,m}^{\mathrm{high}}$: lower/upper freeze-anomaly bounds for month $m$.
- $\sigma_{\Delta T,f,m}$: freeze-anomaly spread parameter for month $m$.
- $\Delta T_{f,m,q}$: sampled anomaly for selected freeze event $q$ in month $m$ (bounded to $[\Delta T_{f,m}^{\mathrm{low}}, \Delta T_{f,m}^{\mathrm{high}}]$).
- $A_{f,m}(i)$: day-of-month freeze-anomaly field; $A_{f,m}(i) = \Delta T_{f,m,q}$ when day $i$ is the $q$-th selected freeze day, and $A_{f,m}(i) = 0$ otherwise.
- $s_{h,m}$, $r_{h,m}$: heat-wave start Gaussian spread and repulsion length scales for month $m$.
- $\mathcal{S}_{h,m}$: set of heat-wave start-day indices selected in month $m$.
- $\mathcal{S}_{h,m}^{(k)}$: heat-wave-start selected set after $k$ recursive selections ($\mathcal{S}_{h,m}^{(0)} = \varnothing$).
- $\mathcal{I}_{h,m}^{(k)}$: heat-wave-start candidates remaining at step $k$, defined by $\mathcal{I}_{h,m}^{(k)} = \{1, \ldots, L_m\} \setminus \mathcal{S}_{h,m}^{(k-1)}$.
- $i_{h,m,k}$: heat-wave start day selected at recursion step $k$.
- $\mathrm{score}_{h,m}^{(k)}(i)$: deterministic ranking objective for heat-wave-start candidate $i$ at step $k$; larger score means better candidate for selection.
- $j$: heat-wave spell index within month $m$, ordered by start day.
- $i_m(d)$: day-of-month index corresponding to day-of-year $d$ in month $m$.

- $\Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}$: lower/upper spell-anomaly bounds (above $\bar{T}_m$) for month $m$, loaded from NOAA-derived monthly spell statistics.
- $\sigma_{\Delta T,h,m}$: configured or fallback standard deviation used for spell-anomaly sampling in month $m$.
- $\Delta T_{h,m,j}$: sampled anomaly assigned to spell $j$ in month $m$, bounded to $[\Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}]$.
- $M_{h,m,j}$: spell-magnitude variable for spell $j$ in month $m$; in code-equivalent notation $M_{h,m,j} \equiv \Delta T_{h,m,j}$.
- $A_{h,m}(i)$: day-of-month spell-anomaly field for month $m$, obtained by allocating each spell magnitude $M_{h,m,j}$ to all days in spell block $\mathcal{B}_{h,m,j}$.
- $g_{\text{std}}(z)$, $G_{\text{std}}(z)$: standard-normal density and CDF used in the explicit truncated-normal equations.
- $\varnothing$: empty set; $A \setminus B$: set difference; $A \cup \{x\}$: add element $x$ to set $A$; $|A|$: set size.
- $D_h(d)$, $D_c(d)$: mode-specific clipped duty fractions.
- $D(d)$ in Eq. (251): active duty fraction for the current mode/day (equal to $D_h(d)$ on heating days or $D_c(d)$ on cooling days).
- $f_{\text{day}}$ in Eq. (244): generic clipped-duty expression; in code this is implemented as `dutyApplied`.
- $\dot{Q}_{\text{load}}$, $\dot{Q}_{\text{nom}}$ in Eq. (244): generic load and nominal-capacity placeholders corresponding to the active mode (heating or cooling).

**Temperature-variable identity and derivation map**

The following temperature symbols are all explicit in this implementation:

- $\bar{T}_m$: monthly Connecticut mean ambient from `climate_data.monthlyMean_C`.
- $\sigma_m$: monthly ambient spread from `climate_data.monthlyStd_C`.
- $T_{\infty,\text{base}}(d)$: regressed daily ambient baseline from periodic Gaussian kernel regression on $\bar{T}_m$.
- $\sigma(d)$: regressed daily spread from the same kernel pass on $\sigma_m$.
- $T_{\infty,\text{freeze}}(d)$: intermediate daily ambient after freeze-day cold overrides.
- $T_m(d)$: month-local daily ambient profile used during event overlays inside month $m$.
- $T_\infty(d)$: final day ambient passed to thermal load and capacity re-solves.
- $T_{h,\text{set}}, T_{c,\text{set}}$: resolved heating/cooling setpoints from the annual config (or inherited model defaults).
- $T_b(d), T_t(d)$: bottom/top bed-node temperatures from the two-node steady solve under $Q_{\text{bed}}(d)$ and $T_\infty(d)$.
- For each heat-wave spell $j$: threshold temperature is $\bar{T}_m + \Delta T_{h,m,j}$, where $\Delta T_{h,m,j}$ is a bounded normal draw.

- NOAA spell-bound files used by config generation:
  - raw daily station rows: `Heat Transfer Study/data/climate/noaa_daily_summaries_USW00014740_2016_`
  - derived monthly freeze bounds: `Heat Transfer Study/data/climate/noaa_freeze_day_anomaly_bounds_`
  - derived monthly spell bounds: `Heat Transfer Study/data/climate/noaa_heatwave_spell_anomaly_boun`
  - extraction utility: `Heat Transfer Study/tools/derive_noaa_heatwave_spell_bounds.py`.

**Direct NOAA-derived inputs vs model-computed variables (explicit provenance)**

To make data provenance explicit, the Section 18 symbols split into two groups.

**A. Direct NOAA-derived reference inputs (loaded from config/file).** These are treated as fixed monthly inputs by the year-round model:

- Statewide monthly climate (NOAA CAG monthly series, 2016–2025):

$$\bar{T}_m = \texttt{climate_data.monthlyMean_C}[m], \qquad \sigma_m = \texttt{climate_data.monthlyStd_C}[m].$$

- Station-based freeze statistics (NOAA daily-summaries station USW00014740, 2016–2025):

$$F_m = \texttt{climate_data.freeze.averageDaysPerMonth}[m],$$

$$\mu_{f,m} = \texttt{climate_data.freeze.meanDayOfMonth}[m],$$

$$M_{f,m} = \texttt{climate_data.freeze.meanAnomalyBelowMonthlyMean_C}[m],$$

$$\Delta T^{\text{low}}_{f,m} = \texttt{climate_data.freeze.minAnomalyBelowMonthlyMean_C}[m], \quad \Delta T^{\text{high}}_{f,m} = \texttt{climate_data.freeze.ma}$$

$$\sigma_{\Delta T,f,m} = \texttt{climate_data.freeze.anomalyStd_C}[m].$$

- Station-based heat-wave timing/count statistics (same station/time window):

$$S_m = \texttt{climate_data.heat_wave.averageSpellStartsPerMonth}[m],$$

$$H_m = \texttt{climate_data.heat_wave.averageDaysPerMonth}[m],$$

$$\mu_{h,m} = \texttt{climate_data.heat_wave.meanStartDayOfMonth}[m].$$

- Station-based heat-wave spell-anomaly bounds (from `noaa_heatwave_spell_anomaly_bounds_USW00014740_2` when `useDerivedSpellAnomalyBounds=true`):

$$\Delta T_{h,m} = \texttt{meanAnomalyAboveMonthlyMean_C}[m],$$

$$\Delta T^{\text{low}}_{h,m} = \texttt{minSpellAnomalyAboveMonthlyMean_C}[m], \quad \Delta T^{\text{high}}_{h,m} = \texttt{maxSpellAnomalyAboveMonthlyMean_C}[m],$$

$$\sigma_{\Delta T,h,m} = \texttt{spellAnomalyStd_C}[m].$$

**B. Model-computed variables (not direct NOAA inputs).** These are produced by equations in Section 18 from the NOAA-derived inputs above:

- kernel-regressed daily fields $T_{\infty,\text{base}}(d)$, $\sigma(d)$,
- allocated integer counts $n_{f,m}$, $n_{s,m}$, $n_{h,m}$,
- selected index sets/lists $\mathcal{S}_{f,m}$, $\mathcal{S}_{h,m}$, and spell blocks $\mathcal{B}_{h,m,j}$,
- sampled spell anomalies $\Delta T_{h,m,j}$,
- final ambient $T_\infty(d)$, duties $D_h(d), D_c(d)$, loads, energy, cost, and water outputs.

So, NOAA provides monthly/statistical inputs; Section 18 equations transform those into daily operational trajectories.

### 18.1 Climate reference values and regression baseline (229, 230)

I began with monthly Connecticut reference values:

- monthly mean temperature $\bar{T}_m$,
- monthly standard deviation $\sigma_m$,
- month lengths from the annual configuration.

I mapped each month to a day-of-year reference day at the month midpoint and evaluated all 365 representative days. For each day $d$, I computed a Gaussian-weighted average:

$$\hat{y}(d) = \frac{\sum_i w_i(d) y_i}{\sum_i w_i(d)}, \qquad w_i(d) = \exp\left[-\frac{1}{2}\left(\frac{\delta_i(d)}{h}\right)^2\right],$$

with circular distance

$$\delta_i(d) = \min(|d - d_i|,\ P - |d - d_i|)\,,$$

where $P = 365$ days and $h$ is the configured bandwidth (default 18 days). I applied this same regression twice:

1. once to monthly means to get daily baseline ambient $T_{\infty,\text{base}}(d)$,
2. once to monthly standard deviations to get daily $\sigma(d)$ for climate-context reporting.

In other words, $y_i$ is not a third intermediary variable family; it is whichever reference-value array is active in the current regression pass.

The circular distance is essential because it treats late December and early January as neighboring points, preventing an artificial year-boundary discontinuity.

### 18.2 Freeze-day allocation and anomaly override (231–234)

This subsection introduces the explicit set-recursion logic used for event-day placement in both 18.2 and 18.3.

The NOAA-derived freeze expectation in each month is fractional (for example, 26.3 days). I convert it to an integer count with largest-remainder allocation:

$$n_{f,m} = \text{LR}(F_m).$$

So month $m$ must receive exactly $n_{f,m}$ freeze days.

**Step 18.2-A: Define the candidate universe and initialization.**

$$\mathcal{U}_m = \{1, 2, \ldots, L_m\}, \qquad \mathcal{S}_{f,m}^{(0)} = \varnothing.$$

Here:

- $\mathcal{U}_m$ is the full set of day-of-month candidates,
- $L_m$ is the number of days in month $m$,
- $\varnothing$ is the empty set.

**Step 18.2-B: Define the remaining candidates at selection step $k$.**

$$\mathcal{I}_{f,m}^{(k)} = \mathcal{U}_m \setminus \mathcal{S}_{f,m}^{(k-1)}, \qquad k = 1, \ldots, n_{f,m}.$$

$\setminus$ means set difference: remove already-selected days.

**Step 18.2-C: Score each remaining candidate.**

$$w_{f,m}(i) = \exp\left[-\frac{(i - \mu_{f,m})^2}{2s_{f,m}^2}\right],$$

$$\text{score}_{f,m}^{(k)}(i) = w_{f,m}(i) \prod_{q \in \mathcal{S}_{f,m}^{(k-1)}} \left[1 - 0.65 \exp\left(-\frac{(i-q)^2}{2r_{f,m}^2}\right)\right], \qquad i \in \mathcal{I}_{f,m}^{(k)}.$$

The factor $w_{f,m}(i)$ prefers days near mean freeze day $\mu_{f,m}$. The product term repels new selections from previously selected days. So $\text{score}_{f,m}^{(k)}(i)$ is a ranking objective, not a probability: at each step, the algorithm chooses the remaining day with the largest score.

**Step 18.2-D: Select one day and update by union.**

$$i_{f,m,k} \in \mathcal{I}_{f,m}^{(k)}, \qquad \text{score}_{f,m}^{(k)}(i_{f,m,k}) = \max_{i \in \mathcal{I}_{f,m}^{(k)}} \text{score}_{f,m}^{(k)}(i),$$

$$\mathcal{S}_{f,m}^{(k)} = \mathcal{S}_{f,m}^{(k-1)} \cup \{i_{f,m,k}\}.$$

$\cup$ means set union: add the newly selected day index. Because $i_{f,m,k} \in \mathcal{I}_{f,m}^{(k)}$, it is not already in $\mathcal{S}_{f,m}^{(k-1)}$. Therefore each update adds exactly one new day, so $|\mathcal{S}_{f,m}^{(k)}| = k$.

After $n_{f,m}$ steps:

$$\mathcal{S}_{f,m} = \mathcal{S}_{f,m}^{(n_{f,m})},$$

so $\mathcal{S}_{f,m}$ contains exactly $n_{f,m}$ distinct freeze days.

**Step 18.2-E: Same logic without set notation (code-equivalent view).** If set notation is unfamiliar, the same selection can be written as an index-list loop:

```
selected = []                              # freeze day indices already chosen
for k in 1..n_f,m:
    best_i = None
    best_score = -infinity
    for i in 1..L_m:
        if i in selected:
            continue
        score = exp(-((i-mu_f,m)^2)/(2*s_f,m^2))
        for q in selected:
            score *= (1 - 0.65*exp(-((i-q)^2)/(2*r_f,m^2)))
        if score > best_score:
            best_score = score
            best_i = i
    selected.append(best_i)
```

This is mathematically identical to the set form:

- "$i \in \mathcal{I}_{f,m}^{(k)}$" means "$i$ not in `selected`",
- "$\mathcal{S}_{f,m}^{(k)} = \mathcal{S}_{f,m}^{(k-1)} \cup \{i_{f,m,k}\}$" means "`selected.append(best_i)`".

**Step 18.2-F: Freeze-magnitude definition and day allocation.** Define freeze-anomaly monthly center:

$$M_{f,m} \equiv \Delta T_{f,m}.$$

**How $M_{f,m}$, bounds, and spread are computed from NOAA data (conceptual derivation):**

- Source rows: NOAA daily-summaries station `USW00014740` over 2016–2025.
- Freeze-day definition: $T_{\min} \leq 32°\text{F}$.
- Daily mean proxy in °C: $T_{\text{day}}^{(r)}$ from $T_{\text{AVG}}$ if present, otherwise $0.5(T_{\max} + T_{\min})$, converted to °C.
- For each freeze day $r$ in month $m$, compute cold anomaly below monthly mean:

$$a_{m,r} = \max\Big(\bar{T}_m - T_{\text{day}}^{(r)},\, 0\Big)\,.$$

- Monthly center is the average freeze-day anomaly:

$$M_{f,m} = \frac{1}{N_{f,m}} \sum_{r=1}^{N_{f,m}} a_{m,r},$$

  where $N_{f,m}$ is the number of freeze days used in month $m$.

- Monthly bounds and spread are:

$$\Delta T_{f,m}^{\text{low}} = \min_r a_{m,r}, \quad \Delta T_{f,m}^{\text{high}} = \max_r a_{m,r}, \quad \sigma_{\Delta T,f,m} = \text{pstdev}(\{a_{m,r}\}_{r=1}^{N_{f,m}}).$$

In this repository, these monthly arrays are precomputed and loaded from:

$$M_{f,m} = \texttt{climate_data.freeze.meanAnomalyBelowMonthlyMean_C}[m].$$

$$\Delta T^{\text{low}}_{f,m} = \texttt{climate_data.freeze.minAnomalyBelowMonthlyMean_C}[m], \quad \Delta T^{\text{high}}_{f,m} = \texttt{climate_data.freeze.maxAr}$$

Define method key $\kappa_f = \texttt{freeze.anomalySamplingModel}$ and deterministic set

$$\mathcal{D}_{\text{det}} = \{\texttt{off}, \texttt{none}, \texttt{disabled}, \texttt{deterministic}, \texttt{deterministic_mean}\}.$$

Define triangular set

$$\mathcal{D}_{\triangle} = \{\texttt{triangular}, \texttt{triangular_mean_matched}, \texttt{triangular_between_monthly_min_max}, \texttt{triangular_mean_mat}$$

$$D_{\triangle} = \mathcal{D}_{\triangle} = \{\texttt{triangular}, \texttt{triangular_mean_matched}, \texttt{triangular_between_monthly_min_max}, \texttt{triangular_me}$$

Define clipped freeze parameters:

$$a_{f,m} = \Delta T^{\text{low}}_{f,m}, \quad b_{f,m} = \Delta T^{\text{high}}_{f,m}, \quad \mu^{\star}_{f,m} = \text{clip}(M_{f,m}, a_{f,m}, b_{f,m}),$$

$$c_{f,m} = \text{clip}\Big(3\mu^{\star}_{f,m} - a_{f,m} - b_{f,m},\ a_{f,m},\ b_{f,m}\Big).$$

Define triangular density on $[a_{f,m}, b_{f,m}]$:

$$t_{f,m}(x) = \begin{cases} \dfrac{2(x - a_{f,m})}{(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & a_{f,m} \le x \le c_{f,m}, \\ \dfrac{2(b_{f,m} - x)}{(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & c_{f,m} \le x \le b_{f,m}, \\ 0, & \text{otherwise.} \end{cases}$$

For each selected freeze event $q = 1, \ldots, n_{f,m}$, allocate event magnitude:

$$\Delta T_{f,m,q} = \begin{cases} \mu^{\star}_{f,m}, & \kappa_f \in \mathcal{D}_{\text{det}}, \\ X_{f,m,q}, \quad X_{f,m,q} = \text{Triangular}(a_{f,m}, c_{f,m}, b_{f,m}), & \kappa_f \in \mathcal{D}_{\triangle} \text{ and } b_{f,m} > a_{f,m}, \\ \mu^{\star}_{f,m}, & b_{f,m} \le a_{f,m}, \\ \mu^{\star}_{f,m} \text{ or legacy bounded-normal draw}, & \kappa_f \notin \mathcal{D}_{\text{det}} \cup \mathcal{D}_{\triangle}, \end{cases}$$

followed by clipping to $[a_{f,m}, b_{f,m}]$. For any $a_{f,m} \le u \le v \le b_{f,m}$,

$$\Pr(u \le X_{f,m,q} \le v) = \int_u^v t_{f,m}(x)\, dx.$$

**Code-equivalent generator for default triangular mode:**

$$\text{Initialize RNG with seed } s_f = \texttt{freeze.anomalyRngSeed}.$$

$$X_{f,m,q} = \text{Triangular}(a_{f,m}, c_{f,m}, b_{f,m}), \qquad X_{f,m,q} \leftarrow \text{clip}(X_{f,m,q}, a_{f,m}, b_{f,m}).$$

Define day-of-month freeze-anomaly allocation (piecewise, no indicator notation):

$$\text{Sort selected freeze days as } s_{f,m,1} < \cdots < s_{f,m,n_{f,m}}.$$

$$A_{f,m}(i) = \begin{cases} 0, & i \notin \mathcal{S}_{f,m}, \\ \Delta T_{f,m,q}, & i = s_{f,m,q} \text{ for some } q \in \{1, \ldots, n_{f,m}\}, \end{cases} \qquad i = 1, \ldots, L_m.$$

So selected freeze days can have different magnitudes (stochastic mode) or identical magnitudes (deterministic mode), depending on $\kappa_f$.

Allocation matters because selection and magnitude are different tasks:

- selection ($\mathcal{S}_{f,m}$) decides *which* days are freeze days,
- allocation ($A_{f,m}(i)$) decides *what anomaly value* each selected day gets.

**Step 18.2-G: Apply freeze override on selected days.** Let $i_m(d) \in \{1, \ldots, L_m\}$ be the day-of-month corresponding to day-of-year $d$ in month $m$. Then

$$T_{\infty,\text{freeze}}(d) = \begin{cases} \min\Big(T_{\infty,\text{base}}(d),\, \bar{T}_m - A_{f,m}(i_m(d))\Big), & i_m(d) \in \mathcal{S}_{f,m}, \\ T_{\infty,\text{base}}(d), & i_m(d) \notin \mathcal{S}_{f,m}. \end{cases}$$

So freeze days are forced colder (or equal) relative to the baseline, never warmer.

**Step 18.2-H: Why allocation matters for implementation.** The override reads a day-level anomaly field $A_{f,m}(i)$, not just one month scalar. This preserves the selected-day pattern and supports both deterministic and stochastic freeze magnitudes.

### 18.3 Heat-wave allocation and anomaly override (235–238)

Section 18.3 uses the same set-recursion logic defined in 18.2. The only change is parameter substitution from freeze-day placement to heat-wave start placement:

$$\{n_{f,m}, \mu_{f,m}, s_{f,m}, r_{f,m}, \mathcal{S}_{f,m}\} \to \{n_{s,m}, \mu_{h,m}, s_{h,m}, r_{h,m}, \mathcal{S}_{h,m}\}.$$

So the meanings of $\setminus$, $\cup$, and "already selected" are identical to 18.2. If preferred, use the exact same code-style loop from Step 18.2-E and replace $(n_{f,m}, \mu_{f,m}, s_{f,m}, r_{f,m})$ with $(n_{s,m}, \mu_{h,m}, s_{h,m}, r_{h,m})$.

**Step 18.3-A: Integer monthly counts.**

$$n_{s,m} = \mathrm{LR}(S_m), \qquad n_{h,m} = \mathrm{LR}(H_m).$$

$n_{s,m}$ is number of spells; $n_{h,m}$ is total heat-wave days.

**Step 18.3-B: Spell lengths.** Let $j = 1, \ldots, n_{s,m}$ index spells in month $m$. Initialize

$$\ell_{h,m,j} = 3, \qquad j = 1, \ldots, n_{s,m},$$

then distribute the remaining days

$$e_m = n_{h,m} - 3n_{s,m}$$

across spells to obtain

$$\boldsymbol{\ell}_{h,m} = (\ell_{h,m,1}, \ldots, \ell_{h,m,n_{s,m}}), \qquad \sum_{j=1}^{n_{s,m}} \ell_{h,m,j} = n_{h,m}.$$

**Step 18.3-C: Start-day set construction (same recursion as 18.2).**

$$\mathcal{U}_m = \{1, \ldots, L_m\}, \qquad \mathcal{S}_{h,m}^{(0)} = \varnothing.$$

For $k = 1, \ldots, n_{s,m}$:

$$\mathcal{I}_{h,m}^{(k)} = \mathcal{U}_m \setminus \mathcal{S}_{h,m}^{(k-1)},$$

$$w_{h,m}(i) = \exp\left[-\frac{(i - \mu_{h,m})^2}{2s_{h,m}^2}\right],$$

$$\mathrm{score}_{h,m}^{(k)}(i) = w_{h,m}(i) \prod_{q \in \mathcal{S}_{h,m}^{(k-1)}} \left[ 1 - 0.65 \exp\left( -\frac{(i-q)^2}{2r_{h,m}^2} \right) \right], \qquad i \in \mathcal{I}_{h,m}^{(k)},$$

where $\mathrm{score}_{h,m}^{(k)}(i)$ is again a deterministic ranking objective (not a probability).

$$i_{h,m,k} \in \mathcal{I}_{h,m}^{(k)}, \qquad \mathrm{score}_{h,m}^{(k)}(i_{h,m,k}) = \max_{i \in \mathcal{I}_{h,m}^{(k)}} \mathrm{score}_{h,m}^{(k)}(i),$$

$$\mathcal{S}_{h,m}^{(k)} = \mathcal{S}_{h,m}^{(k-1)} \cup \{i_{h,m,k}\}.$$

Final selected start set:

$$\mathcal{S}_{h,m} = \mathcal{S}_{h,m}^{(n_{s,m})}.$$

**Step 18.3-D: Enforce non-overlap and month bounds.** Raw starts from Step 18.3-C are projected to feasible starts $s_{h,m,j}$. Using the ordered raw starts $i_{h,m,j}$ $(j = 1, \ldots, n_{s,m})$:

$$e_{h,m,j} = \begin{cases} 1, & j = 1, \\ s_{h,m,j-1} + \ell_{h,m,j-1} + 1, & j \geq 2, \end{cases}$$

$$u_{h,m,j} = L_m - \left( \sum_{p=j}^{n_{s,m}} \ell_{h,m,p} + (n_{s,m} - j) \right) + 1,$$

$$s_{h,m,j} = \mathrm{clip}(i_{h,m,j}, e_{h,m,j}, u_{h,m,j}).$$

Spell $j$ occupies

$$\mathcal{B}_{h,m,j} = \{\, i : \; s_{h,m,j} \leq i \leq s_{h,m,j} + \ell_{h,m,j} - 1 \,\},$$

where $i$ is day-of-month index.

**Step 18.3-E: Spell-magnitude definition and distribution-based allocation.** Define spell magnitude for each spell $j$ as

$$M_{h,m,j} \equiv \Delta T_{h,m,j}.$$

Also define the distribution-method key $\kappa$ from `heat_wave.spellAnomalySamplingModel`. In the implementation:

$$\mathcal{D}_{\mathrm{det}} = \{\texttt{off}, \texttt{none}, \texttt{disabled}, \texttt{deterministic}, \texttt{deterministic_mean}\}.$$

Define bounds and clipped parameters:

$$a_{h,m} = \Delta T_{h,m}^{\mathrm{low}}, \quad b_{h,m} = \Delta T_{h,m}^{\mathrm{high}}, \quad \mu_{h,m}^{\star} = \mathrm{clip}(\Delta T_{h,m}, a_{h,m}, b_{h,m}), \quad \sigma_{h,m}^{\star} = \max(\sigma_{\Delta T,h,m}, \varepsilon).$$

$\varepsilon > 0$ is a small numerical floor used to avoid division by zero.

For each spell $j = 1, \ldots, n_{s,m}$, magnitude allocation follows:

$$M_{h,m,j} = \begin{cases} \mu_{h,m}^{\star}, & \kappa \in \mathcal{D}_{\mathrm{det}}, \\ X_{h,m,j}, \quad X_{h,m,j} \text{ has density } f_{h,m}(x), & \kappa \notin \mathcal{D}_{\mathrm{det}} \text{ and } b_{h,m} > a_{h,m}, \\ \mu_{h,m}^{\star}, & \kappa \notin \mathcal{D}_{\mathrm{det}} \text{ and } b_{h,m} \leq a_{h,m}, \end{cases}$$

$$M_{h,m,j} \leftarrow \mathrm{clip}(M_{h,m,j}, a_{h,m}, b_{h,m}).$$

So each spell receives one scalar magnitude $M_{h,m,j}$; all days in that spell share that same value.

Define day-of-month anomaly allocation:

$$A_{h,m}(i) = \sum_{j=1}^{n_{s,m}} M_{h,m,j}\, \mathbf{1}_{\{i \in \mathcal{B}_{h,m,j}\}}, \qquad i = 1, \ldots, L_m,$$

where $\mathbf{1}_{\{\cdot\}}$ is the indicator function. Because $\mathcal{B}_{h,m,j}$ are non-overlapping by construction, at most one term is active for each $i$.

**Step 18.3-F: Explicit truncated-normal density used when $\kappa \notin \mathcal{D}_{\text{det}}$.** Define bounds and clipped parameters:

$$a_{h,m} = \Delta T_{h,m}^{\text{low}}, \quad b_{h,m} = \Delta T_{h,m}^{\text{high}}, \quad \mu_{h,m}^{\star} = \text{clip}(\Delta T_{h,m}, a_{h,m}, b_{h,m}), \quad \sigma_{h,m}^{\star} = \max(\sigma_{\Delta T,h,m}, \varepsilon).$$

Define standard-normal density and cumulative distribution:

$$g_{\text{std}}(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \qquad G_{\text{std}}(z) = \int_{-\infty}^{z} g_{\text{std}}(t)\, dt.$$

For each spell $j$, $\Delta T_{h,m,j}$ is a random variable on $[a_{h,m}, b_{h,m}]$ with density

$$f_{h,m}(x) = \frac{g_{\text{std}}\left(\frac{x-\mu_{h,m}^{\star}}{\sigma_{h,m}^{\star}}\right)}{\sigma_{h,m}^{\star}\left[G_{\text{std}}\left(\frac{b_{h,m}-\mu_{h,m}^{\star}}{\sigma_{h,m}^{\star}}\right) - G_{\text{std}}\left(\frac{a_{h,m}-\mu_{h,m}^{\star}}{\sigma_{h,m}^{\star}}\right)\right]}, \qquad a_{h,m} \le x \le b_{h,m},$$

and $f_{h,m}(x) = 0$ outside $[a_{h,m}, b_{h,m}]$.

So probabilities are computed by integration:

$$\Pr(u \le \Delta T_{h,m,j} \le v) = \int_u^v f_{h,m}(x)\, dx, \qquad a_{h,m} \le u \le v \le b_{h,m}.$$

**Step 18.3-G: Map allocated spell magnitudes to ambient temperature.** If day-of-month $i_m(d) \in \mathcal{B}_{h,m,j}$, then $A_{h,m}(i_m(d)) = M_{h,m,j}$ and

$$T_{h,m,j}^{\text{th}} = \bar{T}_m + M_{h,m,j},$$

and ambient update

$$T_\infty(d) = \max\Big(T_{\infty,\text{freeze}}(d),\, T_{h,m,j}^{\text{th}}\Big) = \max\Big(T_{\infty,\text{freeze}}(d),\, \bar{T}_m + A_{h,m}(i_m(d))\Big).$$

If $i_m(d) \notin \bigcup_{j=1}^{n_{s,m}} \mathcal{B}_{h,m,j}$, no heat-wave override is applied and

$$T_\infty(d) = T_{\infty,\text{freeze}}(d).$$

After all spells are applied in month $m$, this $T_\infty(d)$ is passed into Section 18.4.

In implementation, random sampling uses a fixed seed (`spellAnomalyRngSeed`) for reproducibility. In this repository state, $\Delta T_{h,m}$, $\Delta T_{h,m}^{\text{low}}$, $\Delta T_{h,m}^{\text{high}}$, and $\sigma_{\Delta T,h,m}$ are NOAA-derived from 2016–2025 daily station data (`USW00014740`) with spell-level anomalies grouped by spell start month.

### 18.4 Daily heating and cooling loads (239, 240)

For each representative day $d$, after final ambient assignment $T_\infty(d)$, I rebuilt the bin thermal-loss model and computed required loads:

$$Q_{\text{req},h}(d) = UA_{\text{bin}} \max(T_{h,\text{set}} - T_\infty(d),\, 0)\,,$$

$$Q_{\text{req},c}(d) = UA_{\text{bin}} \max(T_\infty(d) - T_{c,\text{set}},\, 0)\,.$$

The first equation activates only when ambient is below heating setpoint; the second activates only when ambient is above cooling setpoint.

### 18.5 Capacity re-solve and duty application (241–246)

For each day, I did not reuse fixed design capacities. Instead, I re-solved the selected heating or cooling operating point at that day's ambient:

$$Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d)), \quad P_{h,\text{elec}}(d) = P_h(T_\infty(d)),$$

$$Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d)), \quad P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d).$$

I then computed required duty:

$$D_h^*(d) = \frac{Q_{\text{req},h}(d)}{Q_{h,\text{cap}}(d)}, \qquad D_c^*(d) = \frac{Q_{\text{req},c}(d)}{Q_{c,\text{cap}}(d)},$$

and clipped applied duty to physical bounds:

$$D_h(d) = \min(\max(D_h^*(d), 0), 1), \qquad D_c(d) = \min(\max(D_c^*(d), 0), 1).$$

The generic clipped-duty form $f_{\text{day}}$ in Eq. (244) maps to these mode-specific quantities through the active heating/cooling branch.

The applied bed load is then:

$$Q_{\text{bed}}(d) = \begin{cases} D_h(d)\, Q_{h,\text{cap}}(d), & \text{heating day,} \\ -D_c(d)\, Q_{c,\text{cap}}(d), & \text{cooling day,} \\ 0, & \text{idle day.} \end{cases}$$

If required duty exceeds unity, I preserve $D = 1$ as the applied maximum and record unmet load separately; this is how capacity shortfall is represented without violating equipment limits.

### 18.6 Daily bed temperatures, electricity, cost, and water (247–251)

Given applied $Q_{\text{bed}}(d)$, I solved the same two-node steady bed model used elsewhere:

$$\begin{bmatrix} UA_{b,\infty} + UA_{bt} & -UA_{bt} \\ -UA_{bt} & UA_{t,\infty} + UA_{bt} \end{bmatrix} \begin{bmatrix} T_b(d) \\ T_t(d) \end{bmatrix} = \begin{bmatrix} f_b Q_{\text{bed}}(d) + UA_{b,\infty} T_\infty(d) \\ (1 - f_b) Q_{\text{bed}}(d) + UA_{t,\infty} T_\infty(d) \end{bmatrix}.$$

I computed daily electrical energy by mode:

$$E_h(d) = \frac{P_{h,\text{elec}}(d)\, D_h(d)\, 24}{1000}, \qquad E_c(d) = \frac{P_{c,\text{elec}}(d)\, D_c(d)\, 24}{1000},$$

then

$$E_{\text{day}} = E_h(d) + E_c(d).$$

Daily cost is:

$$C(d) = c_{\text{elec}} E_{\text{day}},$$

and daily water replacement estimate is:

$$m_{\text{water}}(d) = \dot{m}_{\text{water},24h}\, D(d),$$

where $D(d)$ is the active heating or cooling duty on that day.

**Section 18 feed-forward map (explicit variable dependency order)**

**18.1 Climate reference values and regression baseline (229, 230). Inputs:**

- month lengths $\{L_m\}$,
- month-midpoint reference days $\{d_i\}$ (equivalently $\{d_m\}$),
- monthly means $\{\bar{T}_m\}$,
- monthly standard deviations $\{\sigma_m\}$,
- kernel bandwidth $h$.

**Feed-forward chain:**

$$\{\bar{T}_m, d_i, h\} \to \delta_i(d),\ w_i(d) \to T_{\infty,\text{base}}(d),$$

$$\{\sigma_m, d_i, h\} \to \delta_i(d),\ w_i(d) \to \sigma(d).$$

**Outputs passed forward:**

- baseline ambient profile $T_{\infty,\text{base}}(d)$,
- daily spread profile $\sigma(d)$.

**18.2 Freeze-day allocation and anomaly override (231–234). Inputs:**

- expected freeze days $F_m$,
- mean freeze day $\mu_{f,m}$,
- freeze monthly center $M_{f,m}$,
- freeze anomaly bounds $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$,
- freeze anomaly spread $\sigma_{\Delta T,f,m}$,
- baseline ambient $T_{\infty,\text{base}}(d)$ from 18.1.

**Feed-forward chain:**

$$F_m \to n_{f,m} = \text{LR}(F_m) \to \mathcal{S}_{f,m},$$

$\{T_{\infty,\text{base}}(d), \bar{T}_m, M_{f,m}, \Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}, \sigma_{\Delta T,f,m}, \mathcal{S}_{f,m}\} \to M_{f,m},\ A_{f,m}(i) \to T_{\infty,\text{freeze}}(d)$ via the 18.2-G conditional o

**Output passed forward:**

- freeze-adjusted ambient profile $T_{\infty,\text{freeze}}(d)$.

**18.3 Heat-wave allocation and anomaly override (235–238). Inputs:**

- expected spell starts $S_m$,
- expected heat-wave days $H_m$,
- mean start day $\mu_{h,m}$,
- heat-wave mean anomaly $\Delta T_{h,m}$,
- heat-wave anomaly bounds $\Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}$,
- heat-wave anomaly spread $\sigma_{\Delta T,h,m}$,
- freeze-adjusted profile from 18.2.

**Feed-forward chain:**

$$\{S_m, H_m\} \to \{n_{s,m}, n_{h,m}\} \to \mathcal{S}_{h,m} \text{ (using the same recursive set logic as 18.2)},$$

$$\{\Delta T_{h,m}, \Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}, \sigma_{\Delta T,h,m}, \mathcal{S}_{h,m}\} \to \{\Delta T_{h,m,j}\}_{j=1}^{n_{s,m}},$$

$$\{T_{\infty,\text{freeze}}(d), \bar{T}_m, \Delta T_{h,m,j}\} \to T_\infty(d) = \max\Big(T_{\infty,\text{freeze}}(d),\, \bar{T}_m + \Delta T_{h,m,j}\Big) \text{ on spell days } i_m(d) \in \mathcal{B}_{h,m,j}.$$

**Output passed forward:**

- final ambient profile $T_\infty(d)$ used in day-level thermal solves.

**18.4 Daily heating and cooling loads (239, 240). Inputs:**

- final ambient $T_\infty(d)$ from 18.3,
- setpoints $T_{h,\text{set}}$, $T_{c,\text{set}}$,
- day-resolved effective bin conductance terms ($UA_{\text{bin}}$).

**Feed-forward chain:**

$$\{T_\infty(d), T_{h,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},h}(d),$$
$$\{T_\infty(d), T_{c,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},c}(d).$$

**Outputs passed forward:**

- required day loads $Q_{\text{req},h}(d)$, $Q_{\text{req},c}(d)$.

**18.5 Capacity re-solve and duty application (241–246). Inputs:**

- required day loads from 18.4,
- final ambient $T_\infty(d)$ from 18.3,
- selected heating and cooling reference points.

**Feed-forward chain:**

$$\{T_\infty(d), \text{heating reference}\} \to \{Q_{h,\text{cap}}(d), P_{h,\text{elec}}(d)\},$$

$$\{T_\infty(d), \text{cooling reference}\} \to \{Q_{c,\text{cap}}(d), P_{c,\text{elec}}(d)\},$$

$$\{Q_{\text{req},h}(d), Q_{h,\text{cap}}(d)\} \to D_h^*(d) \to D_h(d),$$

$$\{Q_{\text{req},c}(d), Q_{c,\text{cap}}(d)\} \to D_c^*(d) \to D_c(d),$$

$$\{D_h(d), D_c(d), Q_{h,\text{cap}}(d), Q_{c,\text{cap}}(d)\} \to Q_{\text{bed}}(d).$$

**Outputs passed forward:**

- applied bed load $Q_{\text{bed}}(d)$,
- mode duty $D(d)$,
- mode electrical power ($P_{h,\text{elec}}(d)$ on heating days or $P_{c,\text{elec}}(d)$ on cooling days).

### 18.6 Daily temperatures, electricity, cost, and water (247–251). Inputs:

- $Q_{\text{bed}}(d)$, duty, and power from 18.5,
- ambient $T_\infty(d)$ from 18.3,
- two-node conductance parameters and split fraction $f_b$,
- electricity price $c_{\text{elec}}$,
- nominal daily water rate $\dot{m}_{\text{water},24h}$.

**Feed-forward chain:**

$$\{Q_{\text{bed}}(d), T_\infty(d), UA_{b,\infty}, UA_{t,\infty}, UA_{bt}, f_b\} \to \{T_b(d), T_t(d)\},$$

$$\{P_{h,\text{elec}}(d), D_h(d)\} \to E_h(d), \quad \{P_{c,\text{elec}}(d), D_c(d)\} \to E_c(d) \to E_{\text{day}},$$

$$\{E_{\text{day}}, c_{\text{elec}}\} \to C(d),$$

$$\{\dot{m}_{\text{water},24h}, D(d)\} \to m_{\text{water}}(d).$$

**Annual aggregation closure:**

$$\sum_{d=1}^{365} \{E_h(d), E_c(d), C(d), m_{\text{water}}(d)\} \to \{\text{annual energy, annual cost, annual water}\}.$$

### 18.7 Literature-backed microbial heat source $q_{\text{bio}}$ for winter heating

To model partial winter reliance on in-bed microbial activity, I added an explicit volumetric source term $q_{\text{bio}}$ in the bottom-mat reduced heating solver. The source is parameterized from oxygen uptake rate (OUR) and oxygen-to-heat conversion enthalpy, then converted to volumetric units for direct insertion into the bed energy equation.

**Primary source references used in this implementation.**

- Sonowal et al. report dewatered-sludge vermicomposting OUR values of $18.9 \pm 0.5$ mg $O_2$/g VS/day initially, and late-stage values as low as 2.0–3.1 mg $O_2$/g VS/day [R20].
- de Guardia et al. use the biodegradation heat model $H_{\mathrm{BIO}} = r_{O_2} h_{\mathrm{BIO}}$, and report literature $h_{\mathrm{BIO}}$ spans roughly 304–448 kJ/mol $O_2$, with 440 kJ/mol $O_2$ used in their calibrated runs [R19].
- Harper et al. report an empirical oxygen-to-heat slope of about 9.8 MJ/kg $O_2$, equivalent to 313.6 kJ/mol $O_2$, providing a lower empirical conversion reference [R18].
- Gea et al. show OUR is the most reliable process activity indicator under dynamic aeration monitoring, and provide broad OUR ranges across wastes that support dynamic activity modeling rather than static maturity-only values [R21].

**Step 1: OUR time profile at the selected winter operating stage.** I represent process-stage decay from an initial value to a mature value with:

$$\mathrm{OUR}(t_d) = \mathrm{OUR}_{\mathrm{mature}} \; + \; (\mathrm{OUR}_{\mathrm{initial}} - \mathrm{OUR}_{\mathrm{mature}}) \exp(-k_d t_d),$$

where $t_d$ is operating day in days and $k_d$ is in day$^{-1}$. The default $k_d = 0.0353$ day$^{-1}$ is the mean implied by reported 45-day fold reductions from Sonowal et al. [R20].

**Step 2: Convert OUR to heat per unit VS mass.** Using the selected scenario enthalpy $h_{O_2}$ (kJ/mol $O_2$):

$$q_{\mathrm{bio,VS}}\ [\mathrm{W/kgVS}] = \mathrm{OUR}\ [\mathrm{mg}\ O_2/\mathrm{gVS/day}] \cdot \frac{h_{O_2}}{32} \cdot \frac{1000}{86400}.$$

The script exposes scenario bounds:

- conservative: $h_{O_2} = 313.6$ kJ/mol $O_2$ (Harper reference),
- base: $h_{O_2} = 376.8$ kJ/mol $O_2$ (midpoint working value),
- aggressive: $h_{O_2} = 440$ kJ/mol $O_2$ (de Guardia working value).

**Step 3: Convert to volumetric source term for the bed field solve.**

$$q_{\mathrm{bio,vol}}\ [\mathrm{W/m^3}] = q_{\mathrm{bio,VS}} \cdot \rho_{\mathrm{bulk}} \cdot f_{\mathrm{VS,wet}},$$

where $\rho_{\mathrm{bulk}}$ is wet bulk density and $f_{\mathrm{VS,wet}}$ is wet-mass VS fraction.

**Step 4: Apply local temperature response and clipping.** I use a bounded $Q_{10}$ scaling:

$$f_T(T) = \mathrm{clip}\Big(Q_{10}^{(T-T_{\mathrm{ref}})/10}, f_{T,\mathrm{min}}, f_{T,\mathrm{max}}\Big), \qquad q_{\mathrm{bio,local}} = q_{\mathrm{bio,vol}}\, f_T(T).$$

Defaults are $Q_{10} = 2.0$, $T_{\mathrm{ref}} = 22°\mathrm{C}$, and clip 0.35–1.65, preventing runaway positive feedback in the explicit relaxation loop.

**Step 5: Insert into the bed PDE update.** For the 2D reduced field, the interior update adds:

$$\frac{\partial T}{\partial t} \supset \frac{q_{\mathrm{bio,local}}}{\rho c_p},$$

so the implemented source competes directly with evaporation and wall-loss sinks in the same units.

**Step 6: Net winter external-heating requirement.** For duty and feasibility calculations, the winter external requirement is decomposed as:

$$Q_{\text{loss,gross}} = UA\,(T_{\text{set}} - T_{\infty}), \qquad Q_{\text{ext,req}} = \max(Q_{\text{loss,gross}} - Q_{\text{bio,ref}},\, 0)\,,$$

where $Q_{\text{bio,ref}}$ is the temperature-adjusted reference bioheat at $T_{\text{set}}$. This is now printed explicitly in the heating summary as gross loss, microbial credit, and residual external requirement.

### 18.8 Practical interpretation for report writing

The implemented $q_{\text{bio}}$ model is intentionally transparent and auditable:

- every conversion step is explicit (OUR $\rightarrow$ W/kgVS $\rightarrow$ W/m$^3$),
- all literature-sensitive terms are configuration parameters (OUR bounds, $h_{O_2}$ scenario, decay rate, VS fraction),
- solver outputs now separate mat heat and microbial heat contributions,
- the net-load decomposition is reported explicitly, so winter feasibility claims can be traced to assumptions instead of hidden fitting.

### Implementation trace (how I actually executed it in code)

I executed this workflow through the following function chain:

- configuration and climate defaults: `year_round_analysis_config(...)`,
- integer allocation and event placement: `largest_remainder_allocation(...)`, `gaussian_weighted_day_sele` `heatwave_day_mask(...)`,
- spell anomaly sampling and overlays: `sample_triangular_mean_matched(...)` (freeze default), `sample_truncated_normal(...)` (heat-wave and legacy freeze mode), `circular_gaussian_kernel_reg` `build_representative_climate_year(...)`,
- microbial source assembly for winter heating: `_bioheat_base_terms(...)`, `_bioheat_temperature_factor(..` `_bioheat_reference_power_w(...)`,
- daily simulation task: `simulate_year_round_day_task(...)`,
- ambient-specific re-solve of operating points: `simulate_heating_reference_point(...)` and `simulate_summer_cooling_point(...)`,
- bottom-mat field solve with explicit $q_{\text{bio}}$ insertion: `simulate_bottom_mats_state(...)`,
- daily and annual aggregation: `build_year_round_payload(...)`,
- report export: `write_year_round_summary(...)`.

### Reproducibility notes

To ensure reproducibility, I preserved deterministic timing selection rules (fixed score-maximization formulas over remaining start-day candidates), then used seeded bounded-normal sampling for spell magnitudes (fixed `spellAnomalyRngSeed`), plus explicit clipping/guardrails for non-physical values and explicit annual summary totals for freeze days, heat-wave starts, heat-wave days, energy, cost, water, and unmet-load equivalent.

In summary, the Section 18 implementation is an equation-driven annual operations layer with deterministic timing and seeded stochastic spell intensities: smooth baseline climate from monthly reference values, event overlays with bounded spell anomalies, ambient-specific capacity re-solving, physically bounded duty application, and day-level thermo-economic output accumulation.